Business Statistics

Types and Orientation of Data
Flash Cards

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Flash Cards
What data type, view, and format

- Structured or Unstructured?
  - **Structured**: row-column format
- Cross-sectional, time-series, or panel?
  - **Cross-sectional:** Multiple subjects
- Stacked or unstacked
  - **Unstacked:** Wide, single subject per row

Store ID	Location	Sales In May
A	Downtown	42,000
B	Downtown	35,000
C	Airport	58,000
D	Campus	31,000

Flash Cards
What data type, view, and format

- Structured or Unstructured?
  - **Structured:** row-column format
- Cross-sectional, time-series, or panel?
  - **Panel:** Multiple subjects, multiple periods
- Stacked or unstacked
  - **Stacked:** Tall, stores are in multiple rows

StoreID	Month	Sales	Customers
A	Jan	42000	820
A	Feb	39000	780
A	Mar	45000	860
B	Jan	31000	640
B	Feb	29000	610
B	Mar	34000	690

Flash Cards
What data type, view, and format

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  - **Structured:** row-column format

Date	Website Visits	Avg Session Time (min)	Conversions
5/1/2025	420	3.1	18
5/2/2025	395	2.9	16
5/3/2025	510	3.4	24
5/4/2025	475	3.2	21
5/5/2025	530	3.6	27
5/6/2025	498	3.3	23

- Cross-sectional, time-series, or panel?
  - **Time-series:** Single subject, multiple periods

- Stacked or unstacked
  - **Unstacked:** Row has all columns/characteristics for an observation single period on its own row

Flash Cards
What data type, view, and format

- Structured or Unstructured?
  - **Structured:** row-column format
- Cross-sectional, time-series, or panel?
  - **Panel:** If shift is a time period
  - **Cross-sectional:** If shift is a category/team
- Stacked or unstacked
  - **Stacked:** Tall, single observation per row
  - Machines have more than one row for shifts

MachineID	Shift	UnitsProduced	Defects
M1	Day	420	6
M1	Night	395	8
M2	Day	460	5
M2	Night	430	7
M3	Day	480	4
M3	Night	455	6

Flash Cards
What data type, view, and format

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- Cross-sectional, time-series, or panel?

10:42 am – customer: my order still hasn't arrived
10:43 am – agent: sorry about that, can I get your order #
10:44 am – customer: yeah it's 839201
10:47 am – agent: looks like it shipped yesterday

- Stacked or unstacked

Flash Cards
What data type, view, and format

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  - **Unstructured:** not row-column format

10:42 am – customer: my order still hasn't arrived
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Flash Cards
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  - **Structured:** row-column format
- Cross-sectional, time-series, or panel?
  - **Panel:** Multiple stores, multiple years
- Stacked or unstacked
  - **Unstacked:** Wide, single subject per row

StoreID	Sales 2022	Sales 2023	Sales 2024	Employees
A	480,000	510,000	545,000	12
B	390,000	405,000	420,000	9
C	620,000	655,000	690,000	15
D	350,000	365,000	380,000	7

Flash Cards
What data type, view, and format

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- Cross-sectional, time-series, or panel?
- Stacked or unstacked

Store was clean, staff was friendly.
Lines were long!!! waited ~15 minutes 😟
Loved the location but parking was terrible
ok experience. nothing special.

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  - **Structured:** row-column format

- Cross-sectional, time-series, or panel?
  - **Cross-sectional:** Multiple subjects

- Stacked or unstacked
  - **Stacked:** Tall, single observation per row
  - Students have more than one row for exams

StudID	Exam	Score	StudyHours
S01	Midterm	82	14
S01	Final	88	18
S02	Midterm	76	11
S02	Final	81	15
S03	Midterm	91	19
S03	Final	94	22